

Unified Solution of Conjugate Fluid and Solid Heat Transfer–Part II. High-Order Conjugate Heat Transfer

Shu-Jie Li^{1,*} and Lili Ju²

¹ *Division of Mechanics, Beijing Computational Science Research Center, Beijing 100193, China*

² *Department of Mathematics, University of South Carolina, Columbia, SC 29208, USA*

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Abstract. A high-order unified solution method is presented for simulating the coupled fluid flow and heat transfer phenomena in solid materials. The method integrates the energy transfer processes of fluid and solid by transforming the compressible Navier-Stokes equations into a dimensionless system through a double-time-scale approach. A time scaling factor is introduced into the system to bridge and expedite the energy transfer processes between the mediums, thereby accelerating convergence towards a global steady state. To ensure consistent accuracy across material interfaces, high-order formulations are introduced to obtain the interface temperature and heat flux. The effectiveness of the methods are verified and validated through numerical experiments with a three-dimensional discontinuous Galerkin flow solver. Numerical results demonstrate the globally high-order accuracy of the unified method for fluid-solid conjugate heat transfer problems, enabling rapid and robust convergence with large CFL numbers of up to 10^8 .

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1 Introduction

The coupling of fluid flow and heat transfer in solids presents numerous scientific and engineering challenges. Various applications necessitate addressing conjugate heat transfer (CHT) effects, including turbomachinery, heat exchangers, semiconductor devices, and nuclear reactors, among many others. Traditionally, CHT problems are modeled by combining the Navier-Stokes (N-S) equations for fluid flow with the Fourier-Biot (F-B) equation for solid heat transfer [7,21]. This approach usually utilizes separate solvers for each

*Corresponding author.

Emails: shujie@csrc.ac.cn (S. Li), ju@math.sc.edu (L. Ju)

of the physical field with loose coupling through the exchange of boundary conditions across the domain interfaces [27]. However, loose coupling methods often lead to stability constraints on time step size [5,20,22], resulting in robustness issues and massive time steps. Alternatively, fully coupling methods try to combine the fluid and solid governing equations by incorporating interface conditions into a single system. Nonetheless, developing a fully coupling solver by combining the N-S and F-B equations can be as labor-intensive as developing separate codes for fluid and solid applications, since combining the different existing algorithms and solvers is challenging, if not impossible [10,18,24].

In conjugate heat transfer problems involving incompressible flows, two prevalent approaches were proposed to model the solid heat transfer phenomena. The first approach employs the Fourier-Biot (F-B) equation, while the second one utilizes the energy equation of the incompressible Navier-Stokes (N-S) equations. The latter is viable since the incompressible energy equation can be decoupled from the continuity and momentum equations. However, for the compressible N-S equations, the situation is different since the energy equation is coupled with the continuity and momentum equations. Nordström et al [19] presented a similarity condition between the compressible N-S equations and the F-B equation, but the coupling method at the fluid-solid interface still imposes some limitations due to interface stability constraints. To alleviate these stability constraints, we propose a unified solution framework that can fully inherit the computational capabilities of an existing high-order compressible N-S flow solver. In Part I of this work [15], a new high-order solid heat transfer solver has been developed based on the compressible N-S equations. This paper serves as Part II of the work and extends the proposed method to conjugate fluid and solid heat transfer problems.

The remaining parts of this paper are organized as follows: Section 2 presents the theory and equations of the unified solution approach. Section 3 discusses the numerical schemes. Various CHT problems are tested in Section 4. Finally, concluding remarks are given in Section 5.

2 Governing equations

2.1 Nondimensionalization

The compressible N-S equations are employed to describe the coupled fluid flow and solid heat conduction in a unified solution framework. Consider a material with the mass density ρ , temperature T , pressure p , velocity $\mathbf{u}$, total energy E and total enthalpy H per unit mass, the governing equations containing the conservation of mass, momentum and energy, optionally with the gravitational acceleration vector $\mathbf{g}$ and the volumetric heat source Q , are written as: for $t > 0$ and $\mathbf{x} \in \Omega_{\text{fluid}} \cup \Omega_{\text{solid}}$,

$$\begin{cases} \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} = 0, \\ \frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = \frac{\partial \sigma_{ij}}{\partial x_j} + \rho g_i, \\ \frac{\partial(\rho E)}{\partial t} + \frac{\partial(\rho u_j E)}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\sigma_{ij} u_j + k \frac{\partial T}{\partial x_j} \right) + \rho u_i g_i + Q, \end{cases} \quad (2.1)$$

where Ω_{fluid} denotes the fluid domain and Ω_{solid} the solid domain, the stress tensor σ_{ij} is modeled by the Newtonian constitution relation:

$$\sigma_{ij} = \mu(T) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu(T) \delta_{ij} \frac{\partial u_k}{\partial x_k} - p \delta_{ij}. \quad (2.2)$$

The variables of the N-S equations (2.1) are nondimensionalized with the characteristic length L , the density ρ , temperature T , sound speed a , thermal conductivity K and viscosity coefficient μ as follows:

$$\mathbf{x} = \frac{\hat{\mathbf{x}}}{L_r}, \quad L_r = \frac{\hat{L}}{L_m}, \quad T = \frac{\hat{T}}{T_r}, \quad \rho = \frac{\hat{\rho}}{\rho_r}, \quad \mathbf{u} = \frac{\hat{\mathbf{u}}}{a_f}, \quad (2.3a)$$

$$t = \frac{\hat{t}}{t_r}, \quad p = \frac{\hat{p}}{\rho_f a_f^2}, \quad E = \frac{\hat{E}}{a_f^2}, \quad K = \frac{\hat{K}}{K_r}, \quad \mu = \frac{\hat{\mu}}{\mu_r}, \quad (2.3b)$$

where the superscript $\hat{\cdot}$ over a variable denotes the variable has a dimension unit and the subscript r denotes a reference state which are given in Section 2.2. The thermal conductivity K_r and viscosity coefficient μ_r corresponds to the reference temperature T_r . $\hat{L}_r$ is the characteristic reference length, and L_m corresponds to the dimensionless characteristic length of the generated mesh. Finally, four dimensionless numbers are introduced to govern the nondimensionalized system: the *Mach number* in a vector form ($\mathbf{M}$), the *Reynolds number* (Re), the *Prandtl number* (Pr), and the *Péclet number* (Pe) with

$$\mathbf{M} = \frac{\mathbf{u}_f}{a_f}, \quad Re = \frac{\rho_f |\mathbf{u}_f| L_r}{\mu_f}, \quad Pr = \frac{\mu_f c_p}{K_f}, \quad Pe_{s/f} = \frac{a_f L_r}{\alpha_{s/f}},$$

where the thermal diffusivities are defined separately for the fluid and the solid

$$\alpha_f = \frac{K_f}{\rho_f c_p}, \quad \alpha_s = \frac{K_s}{\rho_s c_s}.$$

Thus, a unified dimensionless form of the compressible N-S equations integrating the

fluid flow and solid heat conduction can be written as

$$\begin{cases} \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} = 0, \\ \frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j + p \delta_{ij})}{\partial x_j} = \frac{|\mathbf{M}|}{Re} \frac{\partial}{\partial x_j} \left[\mu(T) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_k}{\partial x_k} \right) \right] + \rho g_i, \\ \frac{\partial(\rho E)}{\partial t} + \frac{\partial(\rho u_j H)}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\frac{|\mathbf{M}|}{Re} \mu(T) u_j \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_k}{\partial x_k} \right) - q_{f/s} \right] + \rho u_i g_i + Q, \end{cases} \quad (2.4)$$

where the fluid heat flux is given by

$$q_f = - \frac{|\mathbf{M}|}{(\gamma-1)RePe_f} \mu(T) \frac{\partial T}{\partial \mathbf{n}}$$

and the solid heat flux is given by

$$q_s = - \frac{\lambda_t}{\gamma(\gamma-1)} K(T) \frac{\partial T}{\partial \mathbf{n}}.$$

In order to bridge and expedite the energy transfer processes between the mediums, a time-scaling parameter λ_t is introduced into the solid heat flux. Time scaling of the nondimensionalized N-S system (2.4) is now controlled by λ_t , which will be discussed in detail in Section 2.2. To close the system, the following nondimensionalized equation of state (EoS) for ideal gas law is used

$$p = \frac{\rho T}{\gamma}, \quad (2.5)$$

where γ is the ratio of specific heats of the fluid. Note that this EoS (2.5) is also applied to the solid in our unified approach.

2.2 Time scaling analysis

In Section 2.1, the fluid and solid energy equations are nondimensionalized using two sets of dimensionless numbers:

- $\forall \mathbf{x} \in \Omega_{\text{fluid}}, \quad K_r = K_f, \quad \rho_r = \rho_f, \quad \mu_r = \mu_f, \quad t_r = \frac{L_r}{a_f},$
- $\forall \mathbf{x} \in \Omega_{\text{solid}}, \quad K_r = K_s, \quad \rho_r = \frac{\rho_s c_s}{c_v}, \quad \mu_r = 0, \quad t_r = \frac{L_r^2}{\alpha_s \lambda_t}.$

With this nondimensionalized system, the fluid and solid fields are allowed to be evolving at different rates in time. The time scaling factor λ_t is used for connecting the distinct time scales of fluid flow and solid heat conduction. The ratio of characteristic time scales of solid and fluid is written as

$$\frac{\delta t_s}{\delta t_f} = \frac{Pe_s}{\lambda_t}. \quad (2.6)$$

Given the Péclet number fixed, the time scaling factor λ_t solely control the time evolution rates of the fluid and the solid. Since usually $\delta t_s \geq \delta t_f$, one comes to the conclusion that λ_t belongs to the interval $[1, Pe_s]$. When λ_t takes Pe_s , we have $\delta t_s = \delta t_f$, indicating the fluid and the solid share the same time scale. On the other hand, if $\lambda_t = 1$, the time evolution rate of the solid field is amplified by a factor of Pe_s in theory, which is typically a large value, e.g. 10^6 . This feature can be utilized to accelerate steady-state CHT computations. The underlying acceleration mechanism can be understood by examining the time step sizes in time-stepping methods. In physics, the convergence rate to the steady state depends on how fast initial transient disturbances propagate out of the computational domains. The total number of time steps required for the disturbances to be driven out of the domain by convection or conduction affects the total computational time, depending on the time step size $\Delta t_{f/s}$. In our method, the time step size with the k -th order DG (DG- P_k) in the fluid domain is defined as

$$\Delta t_f = CFL \frac{h_f}{(2k+1)(|\mathbf{v}_f| + a_f)}, \quad (2.7)$$

and in the solid domain it reads

$$\Delta t_s = \frac{CFL}{2} \frac{h_s^2}{(k+1)^2 \alpha_s}, \quad (2.8)$$

where $h_{s/f}$ denotes the mesh size of the solid/fluid domain and the Courant-Friedrichs-Lewy (CFL) number is the unique number that controls the time step sizes of both the fluid and solid fields. The ratio of propagation rates is controlled by the time scaling factor λ_t . When $\lambda_t = CFL = 1$, the disturbances of the fluid and the solid can propagate to the same dimensionless distance, which is approximately one single cell size per time step. As a consequence, the original slow solid heat conduction requiring many time steps to travel across one single cell distance can now be accomplished in only one time step, significantly speeding up the solid computation as well as the global fluid-solid system. In this paper, $\lambda_t = 1$ is used for accelerating steady convergence in various cases.

2.3 Fluid-solid interface consistency

The heat transfer between fluid and solid occurs at the fluid-solid interface $\partial\Omega^i = \Omega_{\text{fluid}} \cap \Omega_{\text{solid}}$, where the continuity of temperature and normal heat flux must be satisfied, namely,

$$T_s = T_f, \quad K_s \frac{\partial T_s}{\partial \mathbf{n}} = K_f \frac{\partial T_f}{\partial \mathbf{n}}. \quad (2.9)$$

To satisfy this condition, various interface methods have been proposed. Zaguskin and Kudryashov in 1960s [4] and Gile [5] proved that using the temperature for the fluid and passing the heat flux to the solid is conditionally stable in finite difference methods, which belongs to the Dirichlet-Neumann condition. This method has been widely used but suffers from stability constrains due to its interface condition. To alleviate the

stability constraints of the Dirichlet-Neumann condition, Scholl et al. [22] developed a Robin-Robin condition, and Nguyen et al. [20] designed a Dirichlet-Robin condition with adaptive relaxation coefficients, among others. In this work, we propose and validate a symmetric interface condition for high-order methods. We start by analyzing the one-dimensional steady heat conduction equation as a model problem. Firstly, we consider a one-dimensional conjugate heat transfer problem in the interval $[-h, h]$, with the fluid-solid interface located at the origin. Isothermal boundary conditions are imposed at the domain boundaries with temperatures T_s and T_f . The exact solutions of the interface temperature T_i and heat flux q_i is given by

$$T_i = \frac{w_s T_s + w_f T_f}{w_s + w_f}, \quad (2.10a)$$

$$q_i = \frac{w_s w_f}{w_s + w_f} (T_s - T_f), \quad (2.10b)$$

where the symmetric weights are defined as $w_s = K_s/h$, $w_f = K_f/h$. Considering the limiting case $h \rightarrow 0$, the interface temperature T_i is written as

$$T_i = \lim_{h \rightarrow 0} \frac{w_s T_s + w_f T_f}{w_s + w_f} = \frac{K_s T_s + K_f T_f}{K_s + K_f}, \quad (2.11)$$

and the interface heat flux q_i converges to

$$q_i = \lim_{h \rightarrow 0} \frac{w_s w_f}{w_s + w_f} (T_s - T_f) = \frac{K_f \hat{q}_s + K_s \hat{q}_f}{K_s + K_f}, \quad (2.12)$$

where $\hat{q}_{s/f}$ are dimensional variables using the same time coordinate. However, in the double-time-scale coordinates, the heat fluxes q_f and q_s evolve two distinct time coordinates, so the connection between the fluid and the solid heat fluxes has to be built. By matching the normal continuity condition of the fluid heat flux

$$\hat{q}_f = \rho_f a_f^3 q_f, \quad (2.13)$$

and the solid heat flux

$$\hat{q}_s = \frac{\gamma(\gamma-1)K_s T_r}{L_r} q_s, \quad (2.14)$$

the ratio of heat fluxes χ_{sf} can be obtained as

$$\chi_{sf} = \frac{q_s}{q_f} = \frac{\rho_f c_v Pe_s \lambda_t}{\rho_s c_s}. \quad (2.15)$$

Substituting (2.15) to (2.12) yields the heat flux q_i at the fluid or solid side $\partial\Omega_{f/s}^i$

$$q_i = \begin{cases} \frac{q_s \chi_{fs} + \lambda q_f}{1 + \lambda}, & \forall \mathbf{x} \in \partial\Omega_f^i, \\ \frac{q_s + \lambda q_f \chi_{sf}}{1 + \lambda}, & \forall \mathbf{x} \in \partial\Omega_s^i, \end{cases} \quad (2.16)$$

where parameters $\chi_{fs} := \frac{1}{\chi_{sf}}$ and $\lambda := \frac{K_s}{K_f}$ are used to simplify the expressions.

Remark 2.1. The interface temperature (2.11) and heat flux (2.16) are compatible to the classic boundary conditions. The effectiveness of the formulations can be examined in the two limiting cases:

- (i) If $\lambda \rightarrow 0$, one has $T_i \rightarrow T_f$ and $q_i \rightarrow 0$. This implies that the interface is adiabatic which is consistent to the classic adiabatic boundary condition.
- (ii) If $\lambda \rightarrow \infty$, one has $T_i \rightarrow T_s$ and $q_i \rightarrow q_f$. This recovers to the classic Dirichlet-Neumann interface condition given by V. Zaguskin and V. Kudryashov in 1960s [4] and M. Gile in 1997 [5].

Using the interface formulations (2.11) and (2.16), high-order accuracy can be preserved even across the fluid-solid interfaces, yielding global, unified high-order solutions. Numerical validations are conducted in Section 4, which show high accuracy in various cases.

3 Numerical schemes

The compressible N-S equations can be written in a compact form as

$$\mathbf{U}_t + \nabla \cdot \mathbf{F}(\mathbf{U}, \nabla \mathbf{U}) = \mathbf{S}(\mathbf{U}), \quad (3.1)$$

where the flux term contains the contributions from both convection and diffusion

$$\mathbf{F}(\mathbf{U}, \nabla \mathbf{U}) = \mathbf{F}_c(\mathbf{U}) - \mathbf{F}_v(\mathbf{U}, \nabla \mathbf{U}).$$

The conservative vector $\mathbf{U}$, the convection flux $\mathbf{F}_c$, the viscous flux $\mathbf{F}_v$ and the source term $\mathbf{S}$ are written as

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho \mathbf{v} \\ \rho E \end{bmatrix}, \quad \mathbf{F}_c = \begin{bmatrix} \rho \mathbf{v} \\ \rho \mathbf{v} \otimes \mathbf{v} + p \mathbf{I} \\ \rho \mathbf{v} H \end{bmatrix}, \quad \mathbf{F}_v = \begin{bmatrix} 0 \\ \boldsymbol{\tau} \\ \boldsymbol{\tau} \cdot \mathbf{v} - \mathbf{q} \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ 0 \\ Q \end{bmatrix}. \quad (3.2)$$

To solve this convection-diffusion equations, high-order discontinuous Galerkin (DG) methods can be used [6, 23]. In this paper, the unified solution method is implemented into a parallel discontinuous Galerkin flow solver [12, 13, 15–17]. The key of DG discretization to the N-S equations is obtained by subdividing the whole computational domain into a set of elements $\{\Omega_h\}$ and followed by performing an integration by parts [15]

$$\sum \int_{\Omega_h} \psi \frac{\partial \mathbf{U}}{\partial t} d\mathbf{x} + \sum \oint_{\partial \Omega_h} \psi \tilde{\mathbf{F}}(\mathbf{U}, \nabla \mathbf{U}) d\mathbf{S} - \sum \oint_{\Omega_h} \nabla \psi \cdot \mathbf{F}(\mathbf{U}, \nabla \mathbf{U}) d\mathbf{x} = \sum \oint_{\Omega_h} \psi \mathbf{S}(\mathbf{U}) d\mathbf{x}. \quad (3.3)$$

Unlike the finite element methods, the intercell continuity of DG is weakly enforced by the interface flux $\tilde{\mathbf{F}}$ between neighboring elements

$$\tilde{\mathbf{F}}(\mathbf{U}, \nabla \mathbf{U}) = \tilde{\mathbf{F}}_c(\mathbf{U}^-, \mathbf{U}^+) - \tilde{\mathbf{F}}_v(\mathbf{U}^-, \mathbf{U}^+, \nabla \mathbf{U}^-, \nabla \mathbf{U}^+), \quad (3.4)$$

where the inviscid flux $\tilde{\mathbf{F}}_c$ is computed by a Riemann solver, and the viscous flux $\tilde{\mathbf{F}}_v$ is computed with the second approach of Bassi and Rebay (BR2) [1]. The boundary fluxes are also weakly enforced with ghost states determined by the corresponding boundary condition, see [15] for details. The semi-discrete system after the spatial discretization is then solved with time-stepping methods, such as the exponential time integration methods [2, 3, 9, 16] or fully implicit methods [17]. In this paper, the prevalent fully implicit backward Euler method is used [17].

3.1 Preconditioned Roe scheme

The inviscid flux $\tilde{\mathbf{F}}_c$ is computed by a preconditioned Roe scheme. It allows the numerical solver to simulate variable-density flows in low speed regime. The inviscid flux of the original Roe's approximate Riemann solver is defined in terms of the two states across a cell surface $\mathbf{U}_L$ and $\mathbf{U}_R$:

$$\tilde{\mathbf{F}}_c = \frac{1}{2}[\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)] - \frac{1}{2}|A|(\mathbf{U}_R - \mathbf{U}_L). \quad (3.5)$$

The artificial viscosity used in the Riemann solver must be modified to account for the low speed preconditioning. For the standard Roe scheme, the artificial viscosity matrix is defined as

$$|A| = R|\Lambda|R^{-1}, \quad (3.6)$$

where $|\cdot|$ denote the componentwise absolute value of matrix. The matrix R consists of the right eigenvectors and Λ is the corresponding diagonal eigenvalue matrix of the flux Jacobian. The equivalent artificial-viscosity matrix for preconditioning must be $|PA|$ where P is a preconditioning matrix. The eigensystem for PA is different from A since the preconditioning modifies the wave speeds of the traveling disturbances. The modified Roe flux is implemented as follows:

$$\tilde{\mathbf{F}}_c = \frac{1}{2}[\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)] - \frac{1}{2}P^{-1}|PA|(\mathbf{U}_R - \mathbf{U}_L). \quad (3.7)$$

The form of the modified artificial-viscosity matrix is not unique. The matrix of Weiss and Smith [25] is used in this work. The modified flux differs from the original flux in the acoustic corrections which are appended to the average flux. Both the preconditioned inviscid flux and the viscous flux are then applied to all domain faces including interior, boundary and fluid-solid interface. Attention should be paid to the computation of fluid-solid interface flux, which must be designed to satisfy the physical continuity condition. This will be discussed in the following section.

3.2 Fluid-solid interface flux

Since the interface flux is a double-value function at each material side, it has to be computed twice corresponding to the fluid side and the solid side. On either side, the inviscid

flux $\tilde{\mathbf{F}}_c = \tilde{\mathbf{F}}_c(\mathbf{U}^-, \mathbf{U}^g)$ and the viscous flux $\tilde{\mathbf{F}}_v = \tilde{\mathbf{F}}_v(\mathbf{U}^b, \nabla \mathbf{U}^-)$ require ghost states $\mathbf{U}^b$ and $\mathbf{U}^g$. They are obtained separately by satisfying the fluid-solid interface condition on each side.

Fluid Side. On the fluid side of the fluid-solid interface, $\mathbf{U}^-$ corresponds to the fluid state. The ghost state $\mathbf{U}^g$ is a function of the fluid state $\mathbf{U}^-$. It is chosen to ensure the impermeability of solid wall boundaries through using the Riemann solver, which yields

$$\mathbf{U}^g = \left[\rho^-, \rho^- \mathbf{u}^g, \frac{p^-}{\gamma-1} + \frac{1}{2} \rho^- |\mathbf{u}^g|^2 \right]^T, \quad \mathbf{u}^g = -\mathbf{u}^-. \quad (3.8)$$

The no-slip velocity condition and temperature continuity are imposed through the viscous flux $\tilde{\mathbf{F}}_v$ with the Dirichlet states $\mathbf{U}^b$ and $\nabla \mathbf{U}^-$ with

$$\mathbf{U}^b = \left[\rho_b, \mathbf{0}, \frac{p^-}{\gamma-1} + \frac{1}{2} \rho_b |\mathbf{u}^-|^2 \right]^T, \quad \rho_b = \frac{\gamma p^-}{T_i}. \quad (3.9)$$

The proposed formulation of interface temperature T_i , (2.11) is now applied here and then the viscous fluid flux is given by

$$\tilde{\mathbf{F}}_v = [0, \boldsymbol{\tau}^- \cdot \mathbf{n}, (\boldsymbol{\tau}^- \cdot \mathbf{u}^-) \cdot \mathbf{n} - q_i]^T, \quad (3.10)$$

where $\boldsymbol{\tau}^-$ corresponds to the viscous stress obtained by the extrapolation gradient $\nabla \mathbf{U}^-$.

Solid Side. On the solid side of the fluid-solid interface, $\mathbf{U}^-$ corresponds to the solid state. The inviscid normal flux is computed by the same Riemann solver $\tilde{\mathbf{F}}_c = \tilde{\mathbf{F}}_c(\mathbf{U}^-, \mathbf{U}^g)$ with the ghost state given by

$$\mathbf{U}^g = \left[\rho^-, \mathbf{0}, \frac{p^-}{\gamma-1} \right]^T. \quad (3.11)$$

The viscous flux is imposed by setting null contributions to the density and momentum components

$$\tilde{\mathbf{F}}_v = [0, \mathbf{0}, q_i]^T, \quad (3.12)$$

where the interface heat flux q_i is obtained by the interface formulation (2.16). The proposed fluid-solid interface fluxes are found to be successful in terms of accuracy and stability in all the computations.

4 Numerical experiments

In the unified solution method, numerical performance is directly inherited from the original flow solver. This means that the existing numerical features of the original DG solver, such as high-order accuracy and large time stepping capabilities, can be preserved in the new unified solver. In this section, we focus on these numerical features and conduct detailed investigations through three CHT benchmark problems at variable parameter conditions.

4.1 Rotating conjugate heat transfer in a circular annulus

To validate high-order accuracy of the unified solution method, a benchmark problem having analytical solutions is proposed, which can be used as a standard CHT benchmark case for other studies as well. The problem involves conjugate fluid flow and solid heat transfer within a solid circular annulus, where the outer circular boundary is rotating. We contribute the exact solutions to this problem with consideration of the effects of viscous dissipation and variable Brinkman/Prandtl numbers. For simplicity, constant thermal conductivities are assumed. A sample solution configuration of the problem is illustrated in Fig. 1, where isothermal boundary conditions are imposed at the inner and outer boundaries of radius R_i and R_o with temperature T_i and T_o , respectively. The exact solutions are expressed in polar coordinates as follows

$$T_{\text{fluid}}(R) = T_o + (T_o - T_i) \left\{ f_1 \ln \left(\frac{R}{R_o} \right) + Br \left[1 - \left(\frac{R_o}{R} \right)^2 \right] \right\}, \quad (4.1a)$$

$$T_{\text{solid}}(R) = T_i + (T_o - T_i) f_2 \ln \left(\frac{R}{R_i} \right), \quad (4.1b)$$

$$f_1 = \frac{\lambda + Br \left[2 \left(\frac{R_o}{R_m} \right)^2 \ln \left(\frac{R_i}{R_m} \right) + \lambda \beta \right]}{\ln \left(\frac{R_m}{R_i} \right) + \lambda \ln \left(\frac{R_o}{R_m} \right)}, \quad (4.1c)$$

$$f_2 = \frac{1 + Br \left[2 \left(\frac{R_o}{R_m} \right)^2 \ln \left(\frac{R_o}{R_m} \right) + \beta \right]}{\ln \left(\frac{R_m}{R_i} \right) + \lambda \ln \left(\frac{R_o}{R_m} \right)}, \quad (4.1d)$$

where the *Brinkman number* Br , the thermal conductivity ratio λ and the parameter β are given by

$$Br = \frac{\mu R_o^2 \omega_o^2}{K_f (T_o - T_i) \beta^2} = \frac{(\gamma - 1) Pr M_o^2}{\left(1 - \frac{T_i}{T_o}\right) \beta^2}, \quad (4.2a)$$

$$\lambda = \frac{K_s}{K_f}, \quad \beta = 1 - \left(\frac{R_o}{R_m} \right)^2. \quad (4.2b)$$

Note that the Brinkman number can be expressed in terms of the fluid Prandtl number Pr and the tangential Mach number M_o at the outer boundary. The geometric parameters of the problem are given by: the inner radius $R_i = 1$, the middle radius $R_m = 1.5$ and the outer radius $R_o = 2$. The fluid-solid interface located at R_m separates the solid region $R \in [R_i, R_m]$ and the fluid region $R \in [R_m, R_o]$. Numerical computations are carried out using DG- P_3 , Mach number $M_o = 0.05$ and Reynolds number $Re = 10$. The effects of variable Prandtl numbers Pr and thermal conductivity ratios λ are also considered. The computed temperature profiles at $Pr = \{0.7, 10, 10^2\}$ and $\lambda = \{10^{-2}, 10^{-1}, 1, 10, 10^2\}$ are plotted in Fig. 2. It is observed that high Prandtl numbers introduce significant temperature

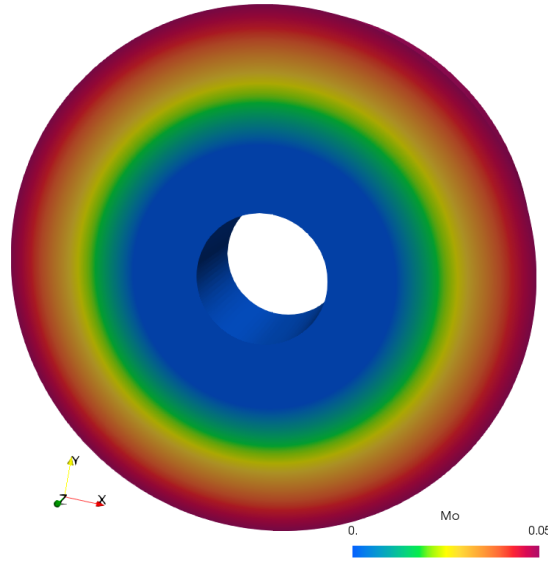


Figure 1: A simulated unified solution of the rotating conjugate heat transfer in a circular annulus at $\lambda = 0.1$, $Pr = 100$, $Mo = 0.05$.

variations due to the heat effect of viscous dissipation. The impact of viscous heating becomes more pronounced from the outer boundary towards the solid wall, attributed to the increasing shear rate. At lower λ values, the maximum temperature occurs at the fluid-solid interface.

To validate high-order accuracy of the proposed method, the numerical orders of accuracy are computed by conducting error convergence study with three sequentially quasi-2D mesh refinements: $20 \times 12 \times 1$, $40 \times 24 \times 1$, and $80 \times 48 \times 1$. Note that the quasi-2D meshes have only one single-layer elements extruded along the z axis. The numerical L_2 and L_∞ errors are used to measure the error convergence rate of temperature and heat flux using $DG-P_3$, which are given in Table 1. We observe that the computed temperature exhibits a fourth-order convergence and the heat flux exhibits a third-order convergence, reaching up to the theoretical orders. The solutions computed on the coarsest mesh ($20 \times 12 \times 1$) are shown in Fig. 2, in which the temperature and heat flux are in excellent agreement with the exact solutions. This case verifies the high-order accuracy of the proposed method for conjugate heat transfer problems, which is barely reported in literature. This exact solution case can be used as a high-order benchmark case of CHT methods.

4.2 Conjugate heat transfer in a square cavity with a heating wall

Next we consider the conjugate heat transfer problem in a square cavity with a heating wall, involving natural convection phenomena driven by gravity and solid heat con-

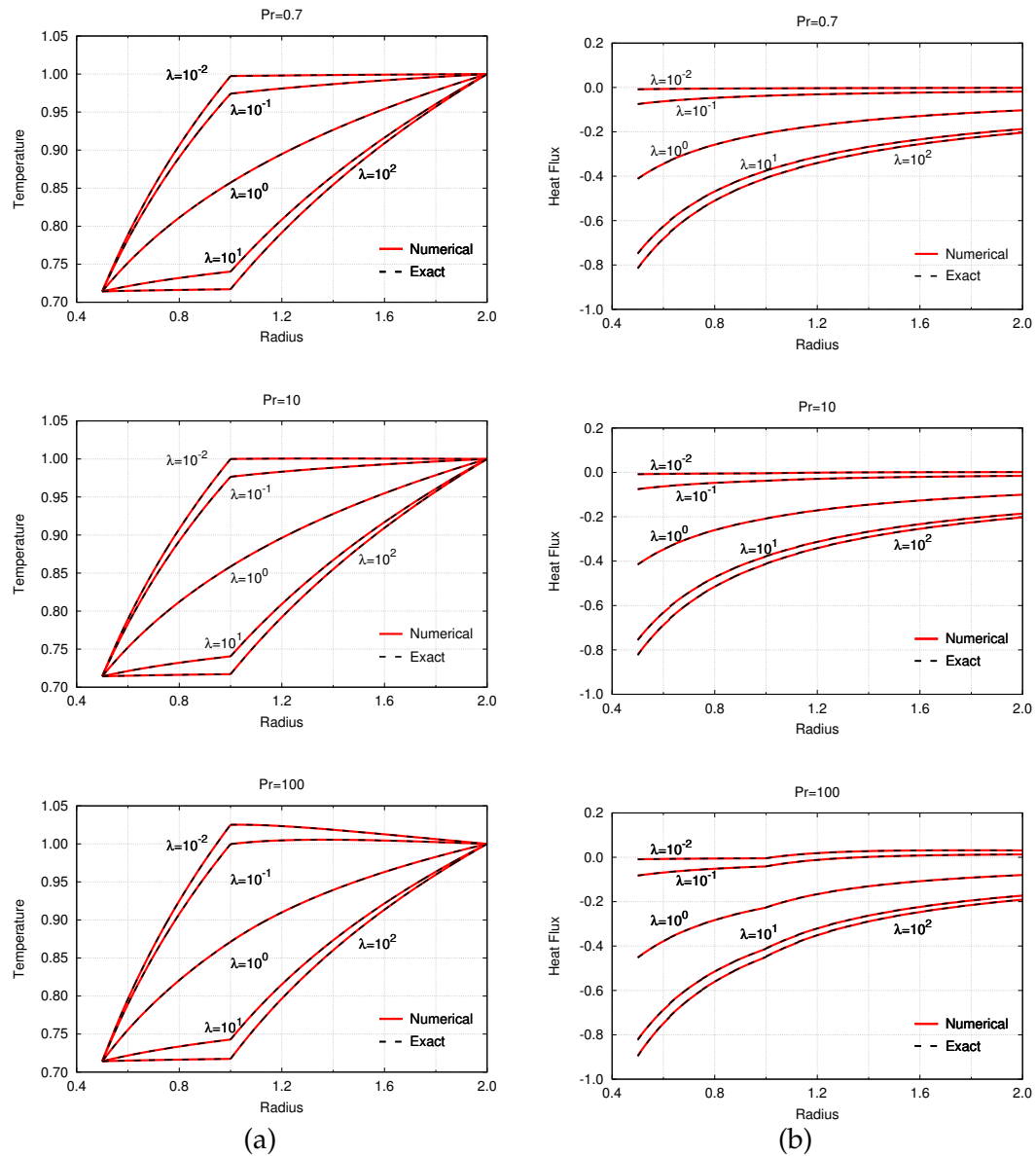


Figure 2: The simulated rotating conjugate heat transfer in a circular annulus at varying Prandtl number Pr and thermal conductivity ratio λ : Comparisons of the numerical solutions of temperature (a) and heat flux (b) with the exact solutions.

duction. The physical pattern is governed by the Prandtl number (Pr) and the Grashof number (Gr), or equivalently, the Rayleigh number ($Ra = Gr Pr$). The Grashof number is

Table 1: The rotating conjugate heat transfer in a circular annulus: Numerical errors and convergence rates.

$\lambda = 10^{-2}$ #element	Temperature				Heat Flux			
	L_2	CR	L_∞	CR	L_2	CR	L_∞	CR
20 × 12	6.191×10^{-6}	n/a	4.820×10^{-5}	n/a	3.291×10^{-6}	n/a	3.232×10^{-5}	n/a
40 × 24	3.921×10^{-7}	3.98	3.317×10^{-6}	3.86	4.128×10^{-7}	2.99	4.320×10^{-6}	2.90
80 × 48	2.694×10^{-8}	3.86	2.486×10^{-7}	3.74	5.305×10^{-8}	2.96	5.770×10^{-7}	2.90
$\lambda = 10^{-1}$								
20 × 12	5.682×10^{-6}	n/a	4.424×10^{-5}	n/a	3.043×10^{-5}	n/a	2.968×10^{-4}	n/a
40 × 24	3.615×10^{-7}	3.97	3.046×10^{-6}	3.86	3.829×10^{-6}	2.99	3.966×10^{-5}	2.90
80 × 48	2.479×10^{-8}	3.87	2.189×10^{-7}	3.80	4.779×10^{-7}	3.00	5.023×10^{-6}	2.98
$\lambda = 10^0$								
20 × 12	5.692×10^{-6}	n/a	2.430×10^{-5}	n/a	1.798×10^{-4}	n/a	1.633×10^{-3}	n/a
40 × 24	3.603×10^{-7}	3.98	1.752×10^{-6}	3.79	2.272×10^{-5}	2.98	2.169×10^{-4}	2.91
80 × 48	2.477×10^{-8}	3.86	1.204×10^{-7}	3.86	2.840×10^{-6}	3.00	2.763×10^{-5}	2.97
$\lambda = 10^1$								
20 × 12	1.000×10^{-6}	n/a	3.739×10^{-5}	n/a	3.789×10^{-4}	n/a	2.969×10^{-3}	n/a
40 × 24	6.496×10^{-7}	3.94	2.395×10^{-6}	3.96	4.732×10^{-5}	3.00	3.966×10^{-4}	2.90
80 × 48	4.187×10^{-8}	3.96	1.520×10^{-7}	3.98	5.892×10^{-6}	3.00	5.023×10^{-5}	2.98
$\lambda = 10^2$								
20 × 12	1.129×10^{-5}	n/a	4.300×10^{-5}	n/a	4.303×10^{-4}	n/a	3.234×10^{-3}	n/a
40 × 24	7.317×10^{-7}	3.95	2.755×10^{-6}	3.96	5.345×10^{-5}	3.01	4.320×10^{-4}	2.90
80 × 48	4.688×10^{-8}	3.96	1.737×10^{-7}	3.99	6.641×10^{-6}	3.01	5.470×10^{-5}	2.98

defined as:

$$Gr = \frac{g\rho_o^2(T_h - T_c)L^3}{\mu_o^2 T_o},$$

where $Pr = 0.71$ is the Prandtl number for air, $g = 9.81$ is the gravitational constant, L represents the height of the cavity, T_h and T_c denote the high and low temperatures applied to the left and right vertical walls respectively. ρ_o and μ_o represent the reference density and viscosity coefficient at the reference temperature $T_o = (T_c + T_h)/2$. The temperature is defined by a dimensionless parameter θ , given by:

$$\theta = \frac{T_h - T_c}{T_h + T_c}.$$

In this setup, the fluid within the cavity is heated along the left wall $T_h = (1 + \theta)T_o$ and cooled along the right wall at $T_c = (1 - \theta)T_o$. The top and bottom boundaries are insulated, and symmetry boundary conditions are imposed on the front and back surfaces of the 3D box. The generated quasi-2D mesh is shown in Fig. 3. Numerical solutions are computed with DG- P_1 at $T_o = 600K$, $\theta = 1/5$, variable thermal conductivity ratios $\lambda = \{1, 5, 10\}$ and Grashof number $Gr = \{10^5, 10^7\}$.

To assess the computational stability and efficiency of the proposed unified solution method, very large time steps are used, exceeding the conventional time-step constraints due to the interface stability condition. The convergence behavior of the method is plot-

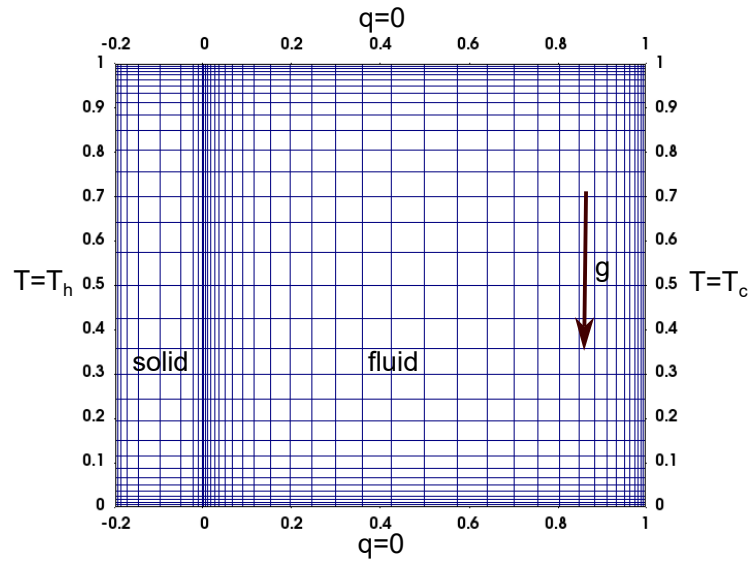


Figure 3: The square cavity with a heating wall: The generated quasi-2D mesh, $\#cell=84 \times 16$.

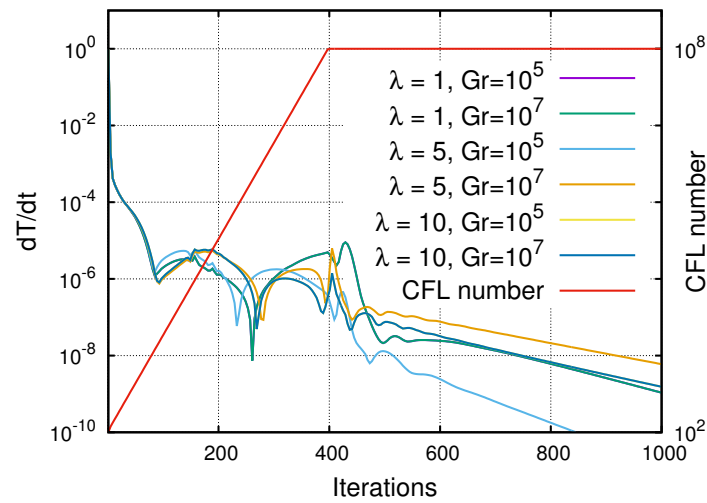


Figure 4: The simulated conjugate heat transfer in a square cavity with a heating wall: Convergence history of the time derivative of temperature (dT/dt) and the CFL number versus the number of iterations.

ted in Fig. 8, where all the solution residuals robustly decrease by ten orders of magnitude with CFL numbers boosted up to 10^8 .

The unified solutions are compared against reference solutions obtained from the works of Hribersek [8] and Kaminski [11]. The comparison, shown in Fig. 6, reveals that our results exhibit good agreement with the previous results. In Fig. 5, it can be observed

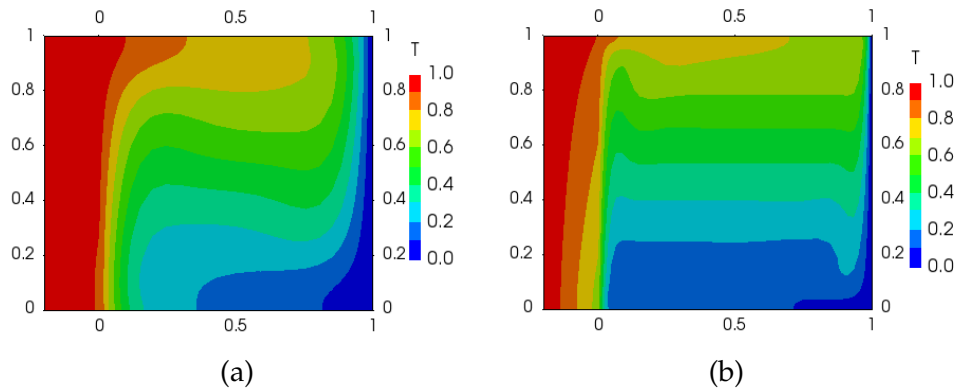


Figure 5: The conjugate heat transfer in a square cavity with a heating wall: Temperature contours at the Grashof number $Gr=10^5$ (a) and $Gr=10^7$ (b) with thermal conductivity ratio $\lambda=10$.

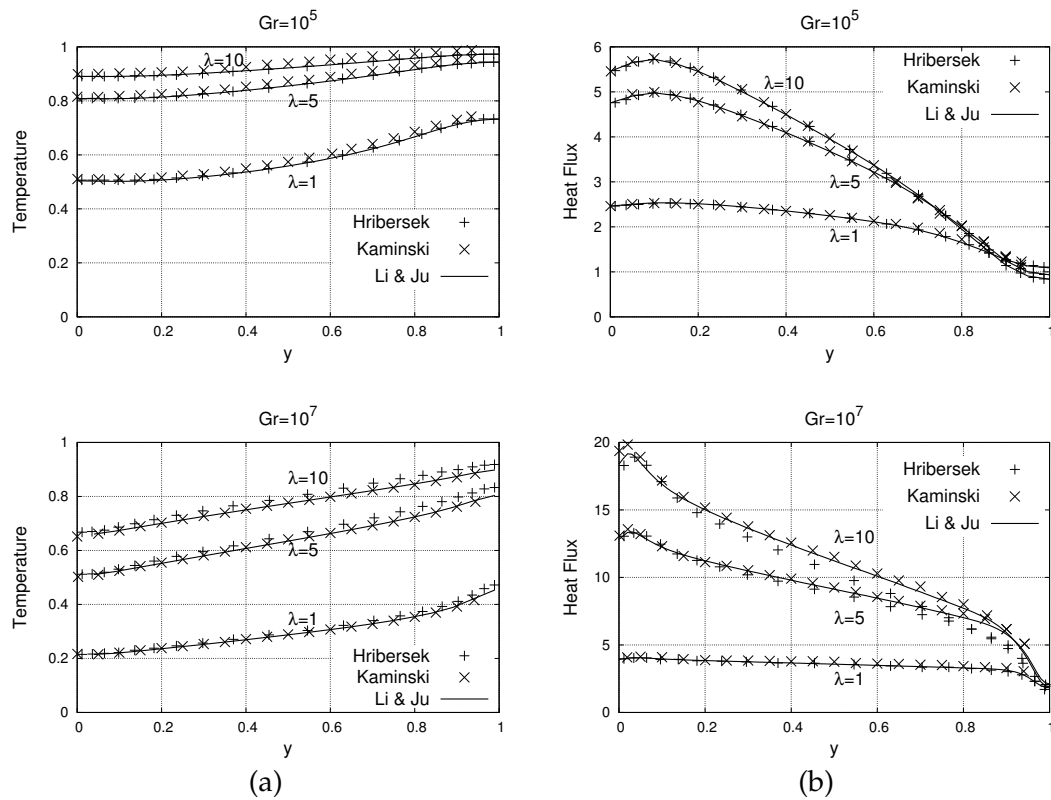


Figure 6: The simulated conjugate heat transfer in a square cavity with a heating wall: Comparison of the simulated temperature (a) and heat flux (b) with the reference results at the thermal conductivity ratio $\lambda = \{1, 5, 10\}$ and the Grashof number $Gr = \{10^5, 10^7\}$. The solutions are taken along the y -direction at location $x=0$.

that the thermal boundary layer tends to become thinner at larger Gr and λ numbers. Overall, the unified solution method is able to deliver good stability and accuracy across a wide range of parameters.

4.3 Natural convection in a 3D finned annulus

The performance for 3D applications of the proposed unified solution method is finally tested through the natural convection in a finned annulus, as illustrated in Fig. 7(a). The annulus consists of two coaxial vertical cylinders and two adiabatic walls on the top and bottom boundaries. A quarter of the annulus is used due to the circumferential symmetry with the symmetric boundary conditions applied to the circumferential end surfaces. The inner and outer radius of the two cylinders are $R_i=1$ and $R_o=2$ with surface temperatures $T_i=1$ and $T_o=0$, respectively. The geometric parameters of the fin are: the thickness $b=L/10$, the radial length $s=L/2$ and the characteristic length $L=1$. The dimensionless numbers of this problem are the ratio of thermal conductivity $\lambda=10$, Prandtl number $Pr=0.71$ and Rayleigh number $Ra=10^4$.

Numerical solutions are computed with DG- P_2 on quadratic curved elements consisting of 5680 hexahedrons and 4020 wedges, as shown in Fig. 7(b). The computed temperature and velocity contours are shown in Fig. 9, where the temperature field is C^0 continuous across the fluid-solid interface, satisfying the physical continuity condition at the interface. The thermal boundary layers due to natural convection appear as expected in the near-wall regions. The computed solutions are compared to those by the FLUENT flow solver in [26] obtained with fine mesh of 1.728 million cells. The comparisons of the

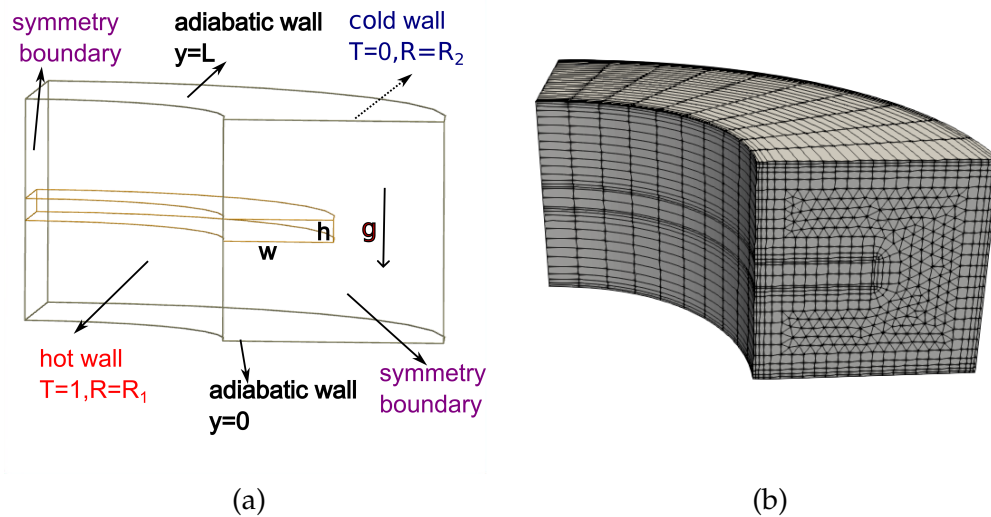


Figure 7: The 3D natural convection in a finned annulus: Geometric configuration (a); generated quadratic curved mesh (b), #hexahedron = 5680, #wedge = 4020.

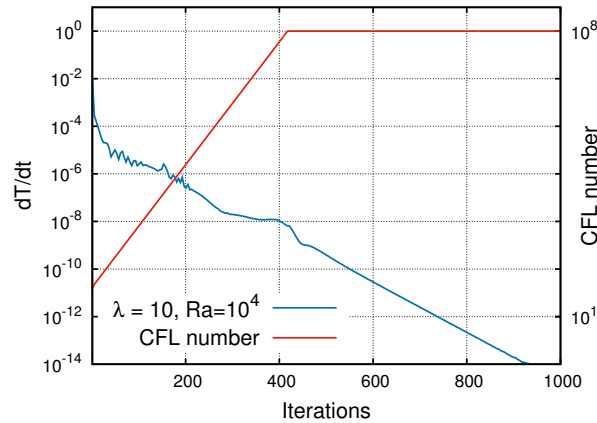


Figure 8: The simulated 3D natural convection in a finned annulus: Convergence history of the time derivative of temperature (dT/dt) and the CFL number versus the number of iterations.

u -velocity profile along the vertical centerline and the v -velocity profile along the horizontal centerline of the $z=0$ plane are shown in Fig. 10(a) with the velocity scaled by the reference velocity $v_r = \alpha_f / L$.

The comparison reveals that the unified fluid-solid solution exhibits good agreement with the FLUENT solver. Convergence history of the unified solution method is plotted in Fig. 10(b), showing great robustness and rapid convergence at large CFL numbers.

5 Conclusions

A three-dimensional unified fluid-solid solution method has been developed to facilitate high-order simulations of conjugate fluid flow and solid heat transfer. The framework unifies the energy transport processes between the fluid and solid domains with a double-time-scale nondimensionalized system, in which an optimal time scaling factor is proposed for convergence acceleration. High-order formulations of the interface temperature and heat flux are proposed in the double-time-scale system, which preserve global high-order accuracy across fluid-solid interfaces. The performance of the unified solution method is demonstrated through its application to three CHT problems using a high-order DG flow solver. An analytical solution was developed for the rotating conjugate heat transfer in a circular annulus problem including the effects of viscous dissipation and variable Brinkman/Prandtl numbers. The exact solution can be used as a benchmark case for numerical validation of other high-order methods as well. Additionally, two natural convection problems are investigated, considering the effects of gravity and solid heat conduction. Numerical results demonstrate globally high-order accuracy of the unified method for fluid-solid conjugate heat transfer problems, enabling rapid and robust convergence with large CFL numbers of up to 10^8 . Future research might explore the integration of more complex physics into the proposed framework.

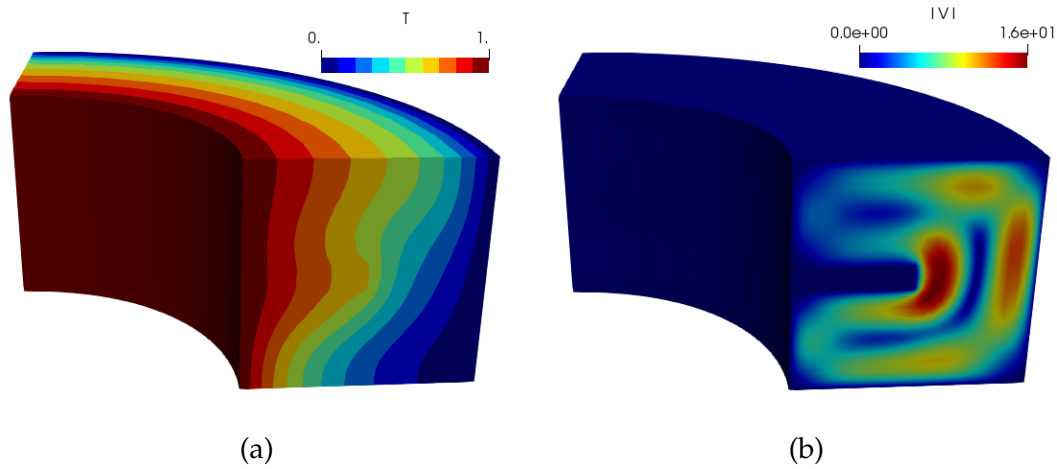


Figure 9: The simulated 3D natural convection in a finned annulus: Temperature (a) ; magnitude of velocity (b).

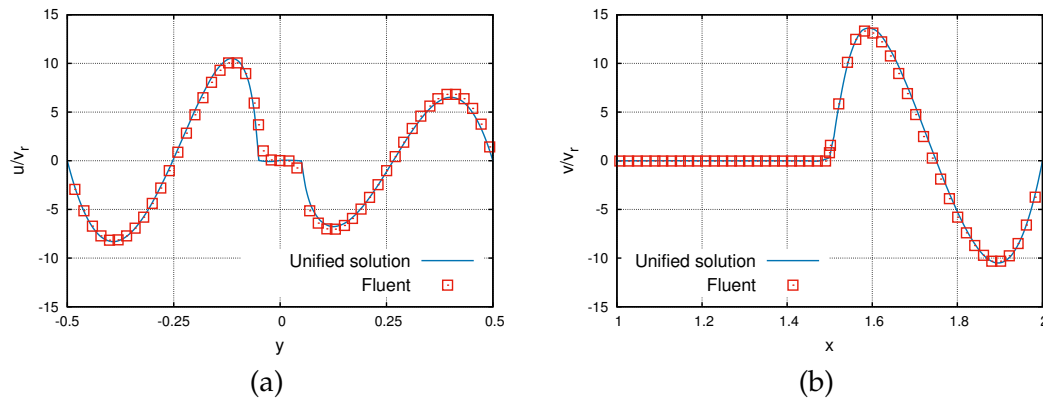


Figure 10: The simulated 3D natural convection in a finned annulus: Comparisons of the u , v velocity profiles with the reference results along the vertical centerline (a) and the horizontal centerline (b) of the $z=0$ plane.

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